Recent Developments in Entropic Gravity: From Microscopic Models to Observational Tests

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***Abstract*—Entropic gravity has evolved from "beautiful but hand-wavy" arguments to explicit microscopic models with testable predictions. This review examines recent developments in 2024-2026, highlighting the transition from philosophical reframings to concrete theoretical frameworks. Key advances include Carney et al.'s quantum microscopic models that derive Newton's law from extremizing free energy, producing observable force noise and decoherence signatures testable in laboratory settings. Bianconi's entropic action principle using quantum relative entropy provides an information-theoretic foundation that recovers General Relativity in appropriate limits. In cosmology, entropic models demonstrate improved observational fits while facing sharper thermodynamic consistency critiques. Wide binary studies continue to probe ultra-weak gravitational regimes with mixed results, while precision spectroscopic measurements challenge both Newtonian and modified gravity interpretations. Verlinde-style emergent gravity shows promise in dwarf spheroidal systems, though cluster-scale tests remain challenging. Recent quantum gravity experiments complicate the traditional "entropic equals classical" narrative, suggesting more nuanced experimental discriminants. This work synthesizes the current landscape, identifying testable predictions that could distinguish entropic gravity mechanisms from conventional gravitational theories.**

***Index Terms*—entropic gravity, emergent gravity, quantum microscopy, thermodynamic cosmology, wide binaries, dark matter alternatives, quantum information, modified gravity, observational tests, force noise**

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I. Introduction

"Entropic gravity" has become an umbrella term for a few related ideas where gravity is **not fundamental**, but instead emerges from **thermodynamics and information** (entropy, horizons, entanglement, coarse-graining). What's changed recently is that parts of the field have shifted from "beautiful but hand-wavy" arguments to **explicit microscopic models** and **clearer observational/experimental discriminants**—alongside sharper critiques of internal consistency.

The field has experienced significant developments in 2024-2026, with researchers moving beyond philosophical reframings toward concrete theoretical frameworks with testable predictions. This transition represents a crucial maturation of entropic gravity from speculative concepts to potentially falsifiable theories.

This review examines the current state of entropic gravity research, focusing on recent breakthroughs in microscopic modeling, cosmological applications, and observational tests. We analyze how these developments address long-standing criticisms while introducing new challenges and opportunities for experimental verification.

II. Background

The concept of entropic gravity emerged from the recognition that gravitational phenomena might arise from thermodynamic principles rather than fundamental force interactions. Early work by Verlinde and others suggested that Newton's laws could be derived from entropic arguments, while subsequent developments explored connections to quantum information theory and holographic principles.

Traditional approaches to entropic gravity faced several challenges: lack of concrete microscopic models, difficulties with observational predictions, and questions about thermodynamic self-consistency. The field was often criticized as providing elegant reframings of known physics without offering genuinely testable alternatives to conventional gravity theories.

Recent developments have begun to address these limitations through multiple parallel advances: explicit quantum mechanical models that predict observable signatures, information-theoretic action principles that recover General Relativity in appropriate limits, and precision observational tests that probe gravitational behavior in previously inaccessible regimes. These advances have transformed entropic gravity from a primarily theoretical exercise into a research program with concrete experimental targets.

III. Methodology

This review systematically examines recent developments in entropic gravity research from 2024-2026, focusing on works that provide explicit theoretical models, observational predictions, or experimental tests. We organize the analysis around several key themes:

A. Microscopic Model Construction

We examine theoretical frameworks that provide explicit quantum mechanical descriptions of entropic force generation, emphasizing models that yield testable predictions for laboratory experiments.

B. Cosmological Applications

We analyze entropic approaches to cosmology, including modified Friedmann equations, dark energy explanations, and observational constraints from multiple datasets including supernovae, baryon acoustic oscillations, and cosmic microwave background measurements.

C. Observational Tests

We review precision tests of gravitational behavior in weak-acceleration regimes, particularly wide binary star systems, dwarf galaxies, and galaxy clusters, examining both supportive evidence and null results.

IV. Results

A. Microscopic Entropic-Force Models

The most significant recent development is the work by **Carney et al.** [1], which builds **explicit quantum microscopic models** where Newton's law arises from **extremizing free energy** of underlying degrees of freedom (e.g., qubits/oscillators), rather than graviton exchange. They present local and nonlocal constructions and—crucially—derive **observable consequences** like **force noise** and **decoherence of spatial superpositions** that could, in principle, be bounded by existing data or near-term experiments.

Quanta's 2025 coverage [2] captures why people paid attention: it's still minority-view and arguably ad hoc, but it's comparatively rare for a quantum-gravity-adjacent idea to produce **lab-facing signatures** rather than only cosmology/astrophysics fits. This upgrades entropic gravity from "philosophical reframing" to "here is a model class with knobs and predictions."

B. Information-Theoretic Action Principles

Bianconi's "Gravity from entropy" [3] proposes an **entropic action** defined as **quantum relative entropy** between the spacetime metric and a matter-induced metric, yielding modified Einstein equations that reduce to General Relativity in a low-coupling regime. The construction introduces a "G-field" as Lagrange multipliers and gives an **emergent small positive cosmological constant** tied to that field.

This represents a broader trend: importing tools from quantum information (relative entropy, density-matrix-like objects) into "emergence of geometry" stories, but in a way that's written down as an **action principle** rather than purely heuristic approaches.

C. Thermodynamic-Cycle Gravity with Lorentz Violations

Isichei & Magueijo [4] reframe General Relativity (and extensions) as a degenerate thermodynamic cycle; adding "work-producing legs" leads to **controlled violations of local Lorentz invariance and energy–momentum conservation** that can drive late-time acceleration.

However, they emphasize a key problem: in the **low-speed/local limit**, these violations look like **drag/energy non-conservation** effects, and in their discussion the **most minimal version is ruled out by Solar-System constraints** unless additional structure (running parameters, preferred-frame subtleties) is added.

D. Entropic Cosmology: Fits and Critiques

Recent entropic cosmology work shows both promising observational fits and fundamental theoretical challenges:

1) Supportive Evidence: **Chagoya et al.** [5] study an entropic cosmological model where a **volumetric term** in the entropy–area relation yields late-time acceleration; they compare to Pantheon supernova data and H(z) measurements, showing an **equivalence to DGP braneworld cosmology**. A related constraints paper [6] derives generalized modified Friedmann equations with multiple entropy terms, claiming good low-redshift agreement and compatibility with SH0ES-like H_0 values.

2) Critical Analysis: **Gohar & Salzano** [7] argue many entropic-cosmology variants are **thermodynamically inconsistent** unless Clausius relations and **mass-to-horizon scaling** are carefully enforced. They claim that once consistency is enforced with matching temperature, many "modified entropy" models become effectively **degenerate** with the original Bekenstein-Hawking setup—and inherit its problems.

A comprehensive 2024 review by **Nojiri, Odintsov & Paul** [8] surveys entropic cosmology built on the **first and second laws** at the apparent horizon, emphasizing model-independent constraints from the second law.

Recent work on **mass-to-horizon entropic cosmology** [9] incorporating Pantheon+, DESI DR2 BAO, CMB distance priors, and growth ($f\sigma_8$) data reports that only **small, near-ΛCDM entropic corrections** are viable; **strong coupling is strongly disfavored**, while weak coupling can be mildly preferred by Bayesian evidence.

E. Ultra-Weak Gravity Tests: Wide Binary Systems

Wide binaries probe accelerations around/below the MOND scale **without dark matter complications**, though systematics are challenging: hidden triples, selection effects, Galactic tides, chance alignments.

A 2024 **MNRAS critical review** [10] emphasizes that different groups reach conflicting conclusions, arguing some "Newton-favoring" analyses may be biased by methodological choices.

Chae (2025) [11] introduces Bayesian 3D orbit modeling including radial velocity and reports **no anomaly at high acceleration** but a **positive anomaly at** $g \lesssim 10^{-9}$ **m/s**^{2}, roughly consistent with MOND expectations.

Pasquini et al. (2026) [12] use **VLT/ESPRESSO** spectra for 26 wide binaries, finding systems that cannot be fit by **bound Newtonian** orbits—while stressing these could be **perturbed/unbound** due to tides or encounters.

Another 2025 analysis [13] argues Newtonian gravity fits better after addressing contamination and modeling choices, demonstrating the debate remains active rather than settled.

F. Cluster and Dwarf Galaxy Tests

Verlinde's "emergent gravity" program aims to reproduce "dark matter" phenomenology from entropy/elastic response ideas. Recent tests show mixed results:

A 2024 cluster study on **SMACS J0723.3-7327** [14] explicitly tests **MOND and Verlinde emergent gravity** using eROSITA and other data, providing another data point in the mixed historical results landscape.

A 2026 preprint [15] compares MOND versus Verlinde emergent gravity in **dwarf spheroidals** and reports preference for the Verlinde model in that dataset, though clusters remain among the hardest environments for "no particle dark matter" explanations.

G. Quantum Gravity Experiments and Entropic Interpretations

A major 2025 Nature paper [16] argues that treating matter fully in **quantum field theory**, some theories with **classical gravity** can still generate entanglement via local processes, with scaling that differs from perturbative quantum gravity. This implies experiments must be designed in parameter regimes that robustly discriminate mechanisms.

This complicates the traditional "entropic equals classical" narrative, making the experimental decision tree more nuanced than "entanglement observed $\Rightarrow$ gravity quantum $\Rightarrow$ entropic dead."

V. Discussion

The current state of entropic gravity research represents a significant maturation from its early, more speculative phases. The most promising recent developments fall into several categories:

Constructive micro-models with testable noise/decoherence signatures (Carney et al.) represent the clearest path toward experimental verification, moving beyond philosophical arguments to concrete laboratory predictions.

Information-theoretic action principles (Bianconi's relative entropy approach) provide mathematical rigor while cleanly recovering General Relativity in appropriate limits, addressing long-standing criticisms about the lack of proper limiting behavior.

Hard-nosed consistency/constraints papers in entropic cosmology expose degeneracies and thermodynamic pitfalls, preventing the field from pursuing attractive but ultimately vacuous modifications.

Precision weak-acceleration tests via wide binaries offer a rapidly evolving observational frontier where data and analysis pipelines continue to improve, though results remain contested.

The field faces several key challenges: entropic mechanisms often flirt with symmetries (Lorentz invariance, equivalence principle) that are extremely tightly tested experimentally. Many entropic cosmology models, while providing good fits to expansion data, struggle with thermodynamic self-consistency when properly analyzed.

However, the transition from "beautiful but hand-wavy" arguments to explicit models with observable consequences represents genuine scientific progress. Even if current models prove incorrect, the methodology of demanding concrete predictions and rigorous theoretical foundations has elevated the field's standards.

The observational landscape remains complex, with different groups reaching conflicting conclusions about the same datasets. This suggests that the field is at a critical juncture where improved observational techniques and more sophisticated analysis methods may soon provide decisive tests.

VI. Conclusion

Entropic gravity has evolved from speculative reframings of known physics to a research program generating testable predictions and confronting observational data. The field's recent developments demonstrate both promise and persistent challenges.

Key advances include: (1) explicit quantum mechanical models producing laboratory-testable signatures, (2) information-theoretic foundations that recover General Relativity in appropriate limits, (3) precision observational tests probing previously inaccessible gravitational regimes, and (4) rigorous theoretical critiques that expose hidden assumptions and degeneracies.

The most immediate prospects for progress lie in: laboratory tests of force noise and decoherence predicted by microscopic entropic models; improved wide binary observations combining Gaia astrometry with high-precision radial velocities; and resolution of thermodynamic consistency issues in entropic cosmology.

Future work should prioritize three directions: **(A)** lab tests of noise and decoherence signatures that could distinguish entropic mechanisms from conventional gravity, **(B)** cosmological model building that addresses both expansion and structure formation while maintaining thermodynamic consistency, and **(C)** precision tests in ultra-weak gravitational regimes that can discriminate between competing theories.

The field stands at a crossroads where theoretical sophistication is meeting observational capability. Whether entropic gravity represents a fundamental reimagining of spacetime or an interesting but ultimately incorrect detour will likely be determined within the next decade through the interplay of improved theory, better observations, and more rigorous experimental tests.

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About This Document

This document represents a technical exploration created through a collaborative process between human and AI. The production process followed these steps:

1) **Discovery**: This review was motivated by recent significant developments in entropic gravity research during 2024-2026, particularly the emergence of explicit microscopic models and laboratory-testable predictions.

2) **Initial Exploration**: Comprehensive survey of recent literature across multiple domains: theoretical quantum mechanics approaches, cosmological applications, observational tests, and experimental proposals.
3) **Synthesis**: The diverse research threads were organized thematically around the progression from microscopic models through cosmological applications to observational tests, structured according to IEEE conference format requirements.
4) **Implementation**: No code implementations were developed for this review, which focuses on theoretical and observational developments.
5) **Visualization**: No additional diagrams were created for this review, as the emphasis is on synthesizing existing theoretical and observational results.
6) **Verification**: All references were scrutinized for authenticity. URLs were tested for accessibility, author names were verified, and content relevance was checked against citations. This verification process is documented in the following section.

This document serves as a comprehensive review of the current state of entropic gravity research, intended for researchers and graduate students working in theoretical gravity, cosmology, and related experimental physics domains. The review emphasizes recent developments that provide testable predictions and observational consequences.

Note on References and Verification

This document contains AI-generated content. All references have been subject to rigorous verification to ensure academic integrity.

Verification Process:

- All URLs were tested for accessibility using automated tools
- Author names were verified against real publications
- DOIs were confirmed where available
- Publication venues (journals, conferences) were validated
- Content relevance was checked against citations

Verification Status in References: Each reference includes a `note` field indicating its verification status:

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- "Standard reference" – Well-known textbook or established work
- "Requires verification" – Needs manual review

Important Notice: Due to the AI-assisted nature of this document's creation, readers should independently verify any references used for critical applications.

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